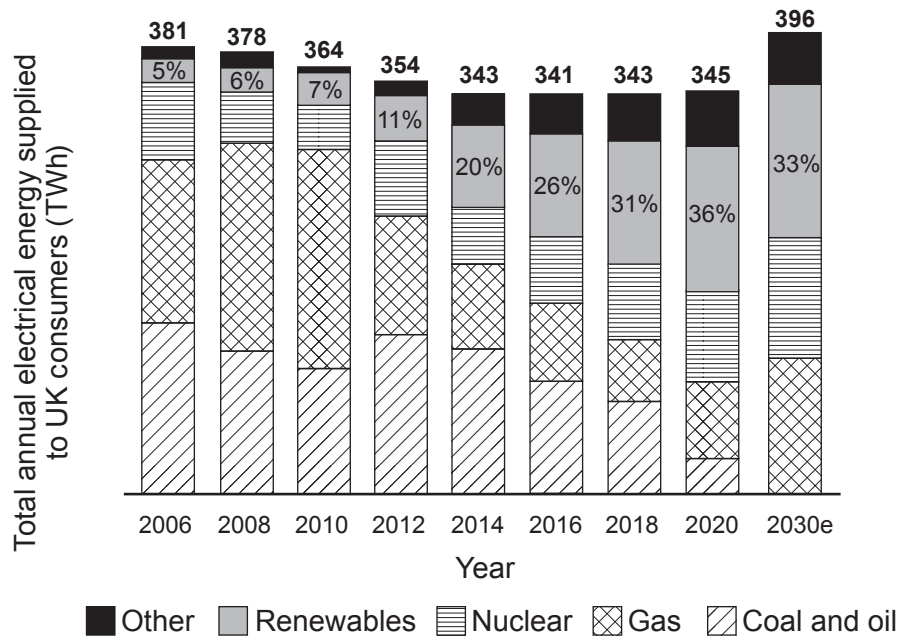


5. The graph below shows the total annual electrical energy supplied to UK consumers and the source of generation.

The total energy is given in terawatt hours (TWh).

The figures for 2030 are estimates.



- (a) Use the graph to answer the following questions.

- (i) Describe how the use of gas has changed between 2006 and 2018. [2]

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.....

.....

- (ii) Calculate how much extra electrical energy will need to be supplied to the UK in 2030 compared to 2020. [1]

extra electrical energy = TWh



- (iii) Calculate how much electrical energy is expected to be generated from renewable sources in 2030. [2]

renewable electrical energy generated = TWh

- (b) One of the largest coal-fired power stations in the UK can generate 950 MW. When it was built the engineers thought it could supply at least 400 000 homes.

The mean power delivered to a house is 1840 W.

Explain whether you agree with the engineers. [3]

(1 MW = 1 000 000 W)

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.....

